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PALAEOMAGNETISM OF UPPER CRETACEOUS VOLCANIC ROCKS IN SICILY

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From Upper Cretaceous volcanic rocks of Southeast Sicily 107 cores from 19 sites were collected giving a mean palaeomagnetic pole position at 62°N, 223°E, A_{95} = 5.4° after AF-cleaning. This pole agrees with the Upper Cretaceous pole of Northern Africa indicating that no large post-Cretaceous relative motion has occurred between Africa and Sicily.

1. Introduction

Palaeomagnetic data for Sicily are lacking except for very young rocks. To facilitate palaeomagnetic comparisons with the Mediterranean area and the adjacent plates a palaeomagnetic survey of Sicily has been carried out. First results from volcanic rocks of Upper Cretaceous age are reported in this paper.

2. Geology

The volcanic formation investigated lies in the extreme Southeast of Sicily at Capo Passero (fig. 1). The rocks which were formed by submarine eruptions consist of compact lava intermixed in some cases with pillows. Into the basaltic mass many dikes were intruded, According to Colacicchi [1] the petrographic composition of the igneous material may be classified as a tephritic magma. The volcanic rocks are overlaid conformally by limestones which are well dated by the aid of fossils as Maestrichtian (Uppermost Cretaceous) [1]. As the deposition of the limestone began shortly after the end of the submarine ejections [1] an Uppermost Cretaceous age (about 70–80 my) can be attributed to the lavas.

Due to several tectonic movements in the Tertiary the igneous rocks were uplifted. A small amount of tilting of the volcanic rocks may be possible but only 2° or less inclining SSE [1; Rittmann, private communication]. Because of its uncertainty and its small influence on the pole position (about 1° in latitude) no bedding correction was applied.

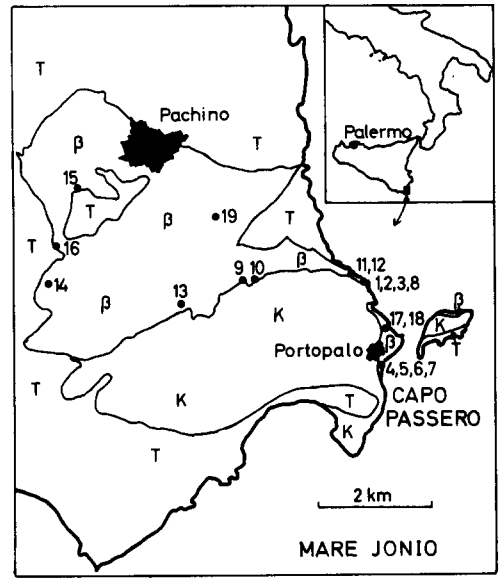


Fig. 1. Outline of the geological map of South East Sicily and sampling sites. β = Upper Cretaceous volcanic rocks; K = Upper Cretaceous (Maestrichtian) limestones; T = Tertiary and younger rocks. Numbers refer to site numbers.

TABLE 1
Site mean palaeomagnetic results after *AF*-cleaning of Capo Passero volcanic rocks, Sicily (location is at 36.7°N, 15.1° E)

Site	*N*	*AF*(Oe)	*RM* 10^{-5} G	*D*	*i*	α_{95}	*k*	Pole position °N	Pole position °E
1	4	400	160	163.7	−13.5	6.4	206	56.6	225.6
2	3	310	140	157.0	+ 5.4	8.6	204	45.1	228.8
3	5	210	140	168.3	− 1.2	10.2	57	52.4	214.4
4	7	300	200	158.4	−26.2	5.1	139	60.0	240.9
5	6	310	390	165.0	−25.8	4.0	274	63.4	229.1
6	7	210	80	172.4	−29.2	5.2	133	67.8	214.9
7	5	140	80	182.1	−38.1	6.4	141	74.5	187.7
8	5	210	620	173.0	−31.1	2.4	998	69.3	214.1
9	7	200	520	161.6	−44.7	2.2	702	71.2	256.8
10	4	350	110	161.4	−33.2	9.2	99	65.2	241.5
11	6	300	380	175.4	−15.3	3.3	412	60.7	204.5
12	4	300	30	163.5	− 4.1	4.8	363	52.1	222.7
13	5	330	380	179.8	−29.6	12.8	36	69.1	195.7
14	5	300	60	166.3	+ 4.8	7.4	106	48.8	216.2
15	5	180	1000	164.5	−23.8	9.5	65	62.1	228.9
16	10	300	70	173.9	−21.1	11.0	20	63.6	208.7
17	6	300	450	165.2	−31.0	4.1	266	66.2	232.3
18	7	250	290	139.5	−39.4	3.2	337	56.7	265.0
19	6	340	40	180.1	−17.5	8.4	63	62.3	194.9
Mean: 19 sites				167.0	−22.0	7.1	22.8	62.1	223.3

N: number of samples, *AF*(Oe): peak value of alternating field cleaning, *RM*: remanent magnetization after alternating field cleaning, *D*: declination, *i*: inclination, α_{95}: radius of the 95% circle of confidence, *k*: precision parameter.

3. Sampling and measurements

Location of the sampling sites are shown in fig. 1. Oriented cores were taken by means of a portable core drilling machine. From each site about 6 different cores (2.5 cm diameter) were collected. A total of 19 sites (107 cores) were sampled.

The remanent magnetization was measured with a fluxgate spinner magnetometer. One pilot specimen from each site was chosen for stepwise demagnetization in alternating magnetic fields using a peak value op to 500 Oe. Most samples carried an unstable component of remanence which disappeared in fields ranging between 100 and 300 Oe for the different sites. All remaining samples were then treated in the appropriate alternating fields. By demagnetization the average initial intensity of magnetization decreased in most cases 20 to 50% at the applied peak alternating field given in table 1. The cleaned remanence directions of samples from each site were

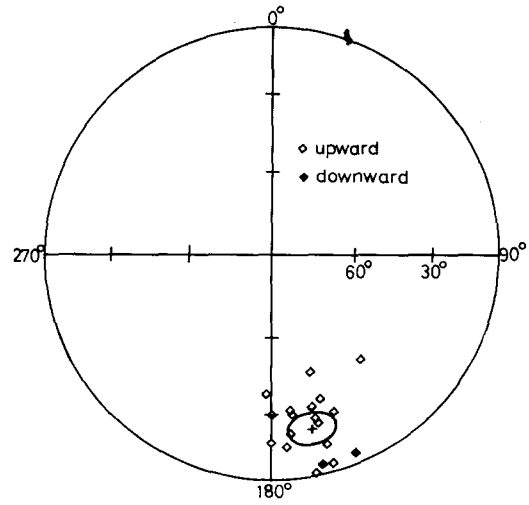


Fig. 2. Stereographic projection of mean site directions after alternating field cleaning. + = Overal mean with the 95% confidence circle (unit weight for each site).

averaged to obtain site mean directions (see table 1). Giving unit weight to each site direction the overall mean direction is $D = 167°$, $i = 22°$ with $\alpha_{95} = 7.1°$ (all sites have reversed magnetization). This yields a pole position at 62.1°N and 223.3°E with $A_{95} = 5.8°$ ($d_p = 4°$, $d_m = 7.5°$). The site mean directions and the overall mean with its circle of confidence are shown in fig. 2.

4. Discussion

In fig. 3 pole positions from Lower Tertiary and Cretaceous for Africa, "stable" Europe, Northern Italy, and Sicily with respect to the present day position of the continents are compiled. The Cretaceous pole for Europe (75°N, 168°E) (Krs [2]) may not be very reliable as it was obtained only from sediments. The only result from volcanic rocks gives a pole not far away at 66°N and 177°E [3], but this result is from one site only. The Early Tertiary pole for "stable" Europe (77°N, 171°E) [4] is near the Cretaceous pole (see fig. 3).

The Eocene pole position for Northern Italy (44°N, 261°E) according to Soffel [5] indicates that Italy rotated counter-clockwise by an angle of 55° between Eocene and Miocene. There is a significant difference between this pole position and the Upper Cretaceous for Sicily. It can be concluded therefore that Sicily was not affected by the large counter-clockwise rotation of Italy during the Tertiary.

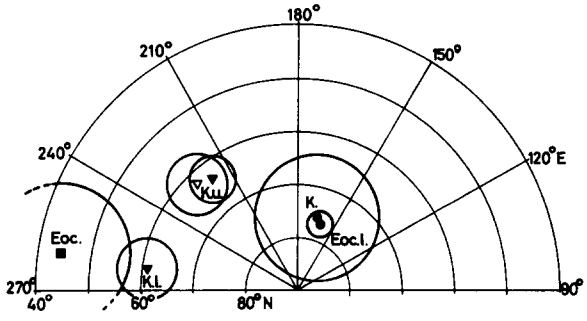


Fig. 3. Compilation of Lower Tertiary and Cretaceous North palaeomagnetic pole positions with their respective circles of confidence. K = Cretaceous; u = upper; l = lower; ● = "stable" Europe (*Eoc l*) [4] and "stable" Europe (*K*) [2]; ■ = Northern Italy [5]; ▼= Africa (*Ku*) [8] and Africa (*Kl*) [7]; ▽ = Sicily, present paper.

As Africa had a large drifting rate during the Cretaceous (e.g. Pitman et al. [6]), pole positions of Upper and Lower Cretaceous should be distinguished. The Lower Cretaceous pole for Africa according to Briden [7] (60.5°N 260.5°E) is significantly different from the Upper Cretaceous pole (63.5°N, 218°E) given by El Shazly and Krs [8]. This pole coincides well with that of Sicily for the Upper Cretaceous (see fig. 3). It is therefore concluded that no large relative motion of Sicily with respect to Africa has occurred at least since the Upper Cretaceous. This is consistent with the suggestion that Sicily is an eastern prolongation of the North African continent (e.g. Caire [9]). The Strait of Sicily seems to be a submerged part of the African continent joining Sicily to North Africa.

The motion of Sicily since the Upper Cretaceous is probably similar to that of Africa with respect to Europe. Following the reconstruction of Pitman et al. [6] the general movement of Africa with respect to "stable" Europe from 80 my to the present is westward (and northward) associated with a counter-clockwise rotation in the order of 20° about a pole in Tangier. This is in agreement with the significant difference of the present day positions of the (Upper) Cretaceous pole positions for Africa and Sicily on one hand and for "stable" Europe on the other.

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References

[1] R. Colacicchi, Geologia del territorio di Pachino (Sicilia meridionale), Geol. Rom. 2 (1963) 343.
[2] M. Krs, Paleomagnetismus (Academia, Praha, 1969).
[3] H. Vollstädt, K. Rother and P. Nozharov, The palaeomagnetic stability and the petrology of some Caenozoic and Cretaceous andesites of Bulgaria, Earth Planet. Sci. Letters 3 (1970) 399.

[4] D.H. Tarling, Paleomagnetic results from the Faeroe Island. In: Palaeogeophysics, ed. by S.K. Runcorn (Academic Press, London, 1970).
[5] H. Soffel, Anticlockwise rotation of Italy between the Eocene and Miocene: Paleomagnetic evidence from the Colli Euganei, Italy, Earth Planet. Sci. Letters 17 (1972) 207.
[6] W.C. Pitman and M. Talwani, Sea-floor spreading in the North Atlantic, Geol.Soc.Amer.Bull. 83 (1972) 619.
[7] J.C. Briden, A new palaeomagnetic result from Lower Cretaceous of East-Central Africa, Geophys.J. 12 (1967) 375.
[8] E.M. El Shazly and M. Krs, Paleography and paleomagnetism of the Nubian Sandstone eastern desert of Egypt. Geol.Rdsch. 62 (1973) 212.
[9] A. Claire, Sicily in its Mediterranean setting. In: Geology and history of Sicily, ed. by W. Alvarez and K.H.A. Gohrbrandt (The Petroleum Exploration Society of Libya, Tripoli, 1970).